

Upper-Layer Applications



Chat Assistants Coding Assistant Other AI Assistant

Application Orchestration

- Prompt Construction
- Retrieval-Augmented Generation
- Tool / API Integration

External Resources



Web Services Data Bases Knowledge Bases

Inference Engine

Request Management

- API & Config
- Queue&Admission
- Batch Scheduling

Model Execution

- Tokenization/ Sampling
- Attention
- FFN / MoE
- Kernels

State & Memory Management

- Model Loading
- KV Cache
- Memory Allocation

Optimization & Compilation

- Quantization
- Graph&Operator Optimization
- Kernel Fusion

Hardware & Distributed Runtime

- Device Backends
- Device Comms
- Parallelism (TP / PP / DP)

Request

Tokens